MIL-A-22397
8 SEPTEMBER 1960
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

ADHESIVE, PHENOL AND RESORCINOL RESIN BASE (FOR MARINE SERVICE USE)

This specification has been approved by the Department of Defense and is mandatory for use by the Department of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.** This specification covers adhesive for bonding wood where an adhesive bond with high strength, resistance to salt water, extreme shrinking and swelling resistance, and long time durability is required. The marine service adhesives are particularly suitable for use in fabricating wood laminates for ship and boat use and for other severe exterior service.

1.2 **Classification.** The adhesive shall be of the following grades as specified (see 6.2):

Grade A — One year storage life.

Grade B — Two months storage life (immediate use).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

PPP-D-729 — Drums, Metal, 55-Gallon (For Shipment of Noncorrosive Materials).

PPP-D-760 — Drums and Pails, Metal (5 and 16.64 Gallons).

STANDARDS

FEDERAL

FED-STD-175 — Adhesives; Methods of Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

37A — Code of Federal Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 38rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. The adhesive furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.1.1 Any changes in the adhesive, such as the adding of pigments, fillers, or diluents, or change in the hardener or the formulation of the base resin will be cause for designating the adhesive as a new material. The changed adhesive shall be given a new manufacturer's designation and request shall be made to the bureau or agency concerned (see 6.3) for qualification testing of the new material.

3.2 Material.

3.2.1 *Resins.* Only resins based on phenol, resorcinol, or a combination of both shall be used in making the adhesive.

3.2.2 *Hardener.* Hardener shall be supplied in either powder or liquid form. Where the hardener is furnished in powder form, it may be combined with part or all of the filler. The hardener shall be mixed with the adhesive prior to application to the joint. Liquid hardener shall be such that it will not settle out during storage between 35° F. and 75° F., to the extent that it cannot be easily brought back into a homogeneous form by slight agitation or mixing.

3.2.3 *Insoluble matter.* The quantity of insoluble matter, including filler and any insoluble partially polymerized resin, shall be not more than 20 percent by weight of the nonvolatile constituents of the adhesive when mixed and ready for use (see 4.6.1).

3.2.4 *Filler.* Where filler is used it shall be an inert insoluble powder incapable of being dissolved in or being swelled excessively by water. The filler may be combined with the resin or supplied separately (see 3.2.3). Amylaceous fillers such as flour or starch, and protein fillers will not be permitted (see 4.6.2).

3.2.5 *pH of set film.* The set adhesive film shall develop a pH value of not less than 3.5 nor more than 11.0 (see 4.6.3).

3.3 Liquid working life. The mixed adhesive shall have a minimum working life of 2.5 hours at 23° ± 1° Centigrade (C.) (73.4 ±1.8° F.) (see 4.6.4).

3.4 Cleanability. The mixed adhesive shall be capable of being easily cleaned from spreading equipment and containers (see 4.6.5).

3.5 Adhesive spread. As mixed for use, the adhesive shall not require a double spread greater than 70 pounds per 1,000 square feet of single glue line to meet the requirements specified in 3.8 and 3.9 (see 3.11(f)).

3.6 Assembly period. When assembled at an ambient room temperature condition of 23° ± 2° C. (73.4° ± 3.6° F.), and using a negligible (less than one minute) open assembly period, and cured in accordance with the cure schedules recommended in the manufacturer's printed instruction, the adhesive shall have a minimum closed assembly period of 15 minutes or less, and the range between the minimum and maximum closed assembly period shall not be less than forty minutes, as determined by meeting the test requirements specified in 3.8 and 3.9.

3.7 **Cure.** When cured in accordance with the cure temperature and cure period combinations recommended in the adhesive manufacturer's printed instruction the adhesive bond shall be such as to meet the requirements specified in 3.8 and 3.9 (see 3.11(i)). In addition when cured at a temperature of 65.5° ± 2° C. (150° ± 3.6° F.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

at the innermost glue line, the adhesive shall require a cure period of not more than 7 hours to meet these requirements.

3.8 Resistance to shear by compression loading in white oak. When tested in accordance with 4.6.6.2, the average percent wood failure for adhesive bonds in any cross section of beam shall not be less than 75 percent, and the average percent wood failure for all cross sections of any beam shall not be less than 80 percent. In addition, the adhesive bonds for all cross sections of any beam shall develop an average block shear strength of not less than 2,000 p.s.i.

3.9 Resistance to delamination in white oak. The total extent of delamination measured on both end surfaces of any specimen shall not exceed 5 percent of the total length of all adhesive lines measured on both end surfaces of that specimen (see 4.6.6.3).

3.10 Storage life (see 4.6.7 and 6.4).

3.10.1 *Grade A.* Grade A adhesive shall be suitable for use and shall meet the requirements of this specification after having been stored in a tightly sealed container for one year from date of manufacture at a temperature of 23° ± 2° C. (73.4° ± 3.6° F).

3.10.2 *Grade B.* Grade B adhesive shall be suitable for use and shall meet the requirements of this specification after having been stored in a tightly sealed container for 2 months from date of manufacture at a temperature of 23° ± 2° C. (73.4° ± 3.6° F).

3.11 Instructions for adhesive use. Unless otherwise specified in the contract or order each unit container of adhesive shall be accompanied by at least one copy of printed instructions, covering at least the following (see 6.2 g):

(a) Identification and date.

(b) Description of the adhesive

(c) Storage care and storage life

(d) Mixing instructions

(e) Useful working life of the mixed adhesive at ambient temperatures of 60° Fahrenheit (F.), 75° F., and 90° F.

(f) Recommended adhesive spread based on double spread in pounds per 1,000 square feet of single glue line (see 3.5)

(g) Recommended glue line pressure

(h) Recommended assembly period schedules as follows (where relative humidity has a critical effect this should be noted):

Ambient assembly temperature (°F.) (air and lumber)	Open assembly period (minutes)	Closed assembly period (minutes)	
		Minimum	Maximum
60	0		
	5		
	10		
	20		
75	0		
	5		
	10		
	20		
90	0		
	5		
	10		
	20		

(i) In a table list recommended cure periods at glue line temperatures of 150° F. (see 4.6.6.1.6), 170° F., and 190° F Also, list the minimum recommended glue line cure temperature which the adhesive will meet all requirements of 3.8 and 3.9 when cured for a period of 24 hours (see 4.6 6 1.6 for cure schedule).

(j) Cleaning instructions.

(k) Health precautions (see 6 6)

3.12 **Label.** A label with the following shall be attached to each unit and shipping container:

(a) Specification No. . .
Grade . .

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

(b) STORE AT A TEMPERATURE BETWEEN 35° F. AND 75° F. KEEP DRY. TIGHTLY COVERED.
DATE OF SHIPMENT
DURING SHIPMENT, AVOID FREEZING TEMPERATURE AND EXTREMES OF HEAT.
KEEP OUT OF SUN
DO NOT USE AFTER . .

(c) FOR USE SEE INSTRUCTION.

(d) CAUTION: MAY CAUSE IRRITATION OF SKIN, EYES, NOSE, AND THROAT.
Avoid prolonged or repeated contact. Avoid breathing dust or prolonged breathing of vapor. Use with adequate ventilation

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 **Qualification.** Qualification shall consist of the procedures specified in 4 2.1 and 4 2.2

4.2.1 *Submittal of information*

4.2.1.1 *Adhesive use instructions and labels* The manufacturer shall submit adhesive use instructions and labels The instructions shall include at least the information outlined in 3 11

4.2.1.2 *Submittal of test data* The adhesive manufacturer shall submit detailed test data which shows that the product will meet the applicable specification requirements

4.2.2 *Qualification test* The qualification test shall consist of the tests specified in 4.6, and any other examination and tests necessary to check conformance with the requirements of this specification. The qualification test including preparation of beams for determining bond quality shall be conducted at a laboratory satisfactory to the Bureau of Ships.

4.3 Lot. A lot shall consist of material from the same batch or blend subjected to the same processing operations and conditions, and submitted for inspection at the same time. A batch shall be defined as that quantity of adhesive which has been subjected to some unit chemical process or physical mixing process intended to make the final product substantially uniform

4.4 Sampling for lot acceptance examination and tests.

4.4.1 *Sampling for examination of filled containers.* A random sample of filled containers shall be selected in accordance with Standard MIL–STD–105 at inspection level I with acceptable quality level equal to 2.5 percent defective, to verify compliance with all stipulations of this specification in regard to fill, closure, labels, instructions, and other requirements not involving tests

4.4.2 *Sampling for acceptance test* Two containers of adhesive shall be selected at random from each lot. From each of these two containers not less than a 1-quart sample and the necessary quantity of hardener shall be taken. The samples shall be composited and redivided into two equal parts; one half shall be used to conduct the tests specified in 4.5.2 and the other half held at a temperature of 23° ± 1° C (73.4° ± 1.8° F.) for referee tests at a laboratory if deemed nec-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

essary by the inspector. Each container shall be properly marked with contract or order number and with such additional information as considered necessary to identify the material which the sample represents. Samples for referee tests need not be held for longer than 60 days. For individual lot orders of 30 gallons or less the procuring activity may waive lot acceptance tests (see 6.2 (f)).

4.5 Lot acceptance examination and tests.

4.5.1 *Examination of filled containers.* Each sample filled container, as specified in 4.4.1 shall be examined for defects of the closure, for evidence of leakage, and for conformance with other requirements not involving tests. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under the required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL-STD-105, the lot represented by the sample shall be rejected.

4.5.2 *Lot acceptance tests.* Where the supplier does not normally conduct tests in accordance with this specification on each batch of his normal commercial production, the adhesive sample selected in accordance with 4.4.2 shall be subjected to the tests specified in 4.6.1, 4.6.2, 4.6.3, 4.6.4, and 4.6.6. Failure of the sample to meet any of the test requirements of 3.2.3, 3.2.4, 3.2.5, 3.3, 3.8 and 3.9 shall result in rejection of the lot. A certification that there is no amylaceous matter will be considered acceptable in lieu of the test specified in 4.6.2. Also, where any test is normally conducted on each batch of adhesive, under the following provisions, it need not be repeated for lot acceptance:

a. The supplier certifies that the lot is part of the batch tested, and the results of test were reasonably accurate, showed specification conformance, and readily available for Government inspection.

4.6 Tests.

4.6.1 *Nonvolatile and insoluble content.*

4.6.1.1 *Nonvolatile content.* Mix at least 100 grams of adhesive according to the manufacturer's directions, weighing to the nearest 0.01 gram. Place approximately 10 grams of fine oven-dried quartz sand in a Petri dish (100 x 22 mm. or its equivalent) together with a small glass stirring rod, and weigh the dish and its contents to the nearest milligram on an analytical balance. Add approximately 2 grams of the mixed adhesive and weigh again to the nearest milligram Mix the adhesive and the quartz sand intimately by means of the stirring rod and distribute uniformly over the bottom of the dish in as thin a layer as possible. Heat the dish, rod, and contents overnight in an oven maintained at 105° ± 1° C. (221° ± 1.8° F.) cool, reweigh, and calculate the percent nonvolatile as follows:

$$\text{Percent nonvolatile} = \frac{\text{(weight of adhesive-loss in weight)}}{\text{weight of adhesive}} \times 100$$

4.6.1.2 *Insoluble matter.* Make this determination on the same batch of mixed adhesive used for determination of nonvolatile content (see 4.6.1.1).

4.6.1.2.1 *Phenol and resorcinol resin.* Weigh approximately 2 grams of the freshly mixed adhesive to the nearest milligram in a 100-milliliter (ml.) beaker or Erlenmeyer flask Add 50 ml. of absolute methyl alcohol. Stir for 3 minutes, then filter through a previously weighed sintered glass crucible of medium porosity. Transfer all residue to the filter with a second 50-ml. portion of methyl alcohol. Wash the residue in the crucible with at least 50 ml. of boiling distilled water Dry to approximately constant weight in an oven maintained at 105° ± 1° C. (221° ± 1.8° F) ap-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

proximately 2 hours. Cool in a dessicator, and weigh to the nearest milligram. Calculate the insoluble content as a percent of the nonvolatile content as follows:

$$\text{Percent insoluble} = \frac{\text{weight of residue x 100}}{\text{weight of adhesive x percent nonvolatile}} \times 100$$

4.6.1.2 *Insoluble matter*. Make this determination on the same batch of mixed adhesive used for determination of nonvolatile content (see 4.6.1.1). Weigh approximately 2 grams of the freshly mixed adhesive to the nearest milligram in a 100-milliliter (ml.) beaker or Erlenmeyer flask. Add 50 ml. of absolute methyl alcohol. Stir for 3 minutes, then filter through a previously weighed sintered glass crucible of medium porosity. Transfer all residue to the filter with a second 50-ml. portion of methyl alcohol. Wash the residue in the crucible with at least 50 ml. of boiling distilled water. Dry to approximately constant weight in an oven maintained at 105° ± 1° C (221° ± 1.8° F.) approximately 2 hours. Cool in a dessicator, and weigh to the nearest milligram. Calculate the insoluble content as a percent of the nonvolatile content as follows:

$$\text{Percent insoluble} = \frac{\text{weight of residue x 100}}{\text{weight of adhesive x percent nonvolatile}} \times 100$$

4.6.2 *Amylaceous matter*. To approximately ½ gram of mixed adhesive add approximately 5 ml. of 95 percent ethyl alcohol in a test tube of small diameter. Shake thoroughly, allow matter to settle, and decant the supernatant liquid. Repeat this process twice with alcohol and three times with distilled water. To the residue after the final decantation, add a drop of a solution made of 5 grams of iodine, 5 grams of potassium iodide, and 100 ml. of distilled water. The formation of a dark reddish or bluish color denotes the presence of amylaceous matter.

4.6.3 *pH*. The adhesive shall be prepared and tested for pH in accordance with method 4011 of Standard FED-STD-175. Where previous tests have shown that pH of the set adhesive film did not fall outside the limits of 5 and 8, for lot acceptance, pH may be determined on the mixed liquid adhesive in lieu of the set film.

4.6.4 *Liquid working life*. Pour approximately 200 grams of freshly mixed adhesive into a large glass test tube or beaker approximately 35 millimeters (mm.) in diameter provided with a glass stirring rod, and immerse the tube with contents in a water bath held at 23° ± 1° C (73.4° ± 1.8° F.). Stir the adhesive frequently with the rod, and note the time at which the adhesive is considered to be definitely no longer spreadable on a wood surface with stiff bristle brush. The time to reach this consistency after mixing is the liquid working life of the adhesive at 23° ± 1° C. (73.4° ± 1.8° F.). In the case of doubt as to whether the material is brushable, a Gardner mobilometer or equivalent (such as a Brookfield Synchro electric Viscometer) shall be filled with the mixed adhesive to within ½ inch of the top. Using the 51-hole disk with a 400-gram total weight, three test runs shall be made, determining in each instance the time in seconds required for the plunger to fall between the 10 centimeter marks. The two readings in nearest conformance shall be averaged and the viscosity calculated in accordance with the following:

Viscosity (poises) = T x 4 x C x K
where

T = time in seconds for 10 centimeter fall

4 = weight, factor instrument calibrated at 100 grams

C = correction factor for the individual instrument (see manufacturer's instructions)

K = constant which is 0.2052

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

If the viscosity, when determined by this method, exceeds 200 poises the material shall be considered no longer brushable.

4.6.5 *Cleanability*. A portion of the mixed adhesive shall be placed in a glue pot and, before the expiration of the working life and at the proper temperature of use as furnished by the manufacturer, shall be completely cleaned from the pot by water, ½ percent caustic solution, alcohol, or a mixture of these.

4.6.6 *Tests for bonds in white oak*. Tests for adhesive bonds in white oak shall consist of shear by compression loading, and cyclic delamination.

4.6.6.1 *Preparation of test beams*.

4.6.6.1.1 *Wood*. Each beam shall consist of six pieces of white oak heartwood ¾ inch thick, 6¼ inches wide, and 40 inches long. The wood shall be free of all defects, and have a slope of grain not steeper than 1 inch rise in 15 inches of length measured on any face or edge. Specific gravity shall be not less than 0.67 based on oven-dry weight and volume. Moisture content of the wood shall be 12 ± 2 percent, and shall be free of case-hardening, collapse, splits, checks, and other drying defects.

4.6.6.1.2 *Machining*. The wood shall be accurately machined to a total thickness tolerance of .01 inch within 24 hours of gluing

4.6.6.1.3 *Adhesive spread*. The adhesive shall be applied uniformly to both faces of each lamination, except that the adhesive shall be omitted from the outer faces of the assembly. The spread shall be the minimum recommended in the manufacturer's printed instruction (see 3.5).

4.6.6.1.4. *Assembly*

4.6.6.1.4.1 *Straight beams*. Two straight beams shall be assembled at an ambient temperature of 23° ± 2° C. (73.4° ± 3.6° F.). Using a negligible open assembly period (close to zero minute) fabricate one beam at the minimum closed assembly period recommended in the manufacturer's printed instruction, and one at the maximum recommended closed assembly period (see 3.6 for assembly period limitations).

4.6.6.1.5 *Glue line pressure*. Throughout the cure cycle, the glue line pressure shall be that recommended in the manufacturer's printed instruction with a tolerance of ± 25 p.s.i., but not to exceed a maximum of 225 p.s.i.

4.6.6.1.6 *Cure*. Cure for beams shall be of a definite schedule, which consists of a heating period, cure period, and cooling period as defined in (a), (b), and (c). During the cure cycle, the ambient curing chamber temperature shall at no time exceed the specified glue line curing period temperature by more than 25° F., nor shall it exceed 215° F. Also, for the cure cycle, the ambient chamber temperature during the heating and cooling periods shall be raised and lowered at approximately 70° F. per hour. Throughout the cure cycle, relative humidity shall be maintained in the curing chamber so as to maintain an equilibrium moisture content of 12 ± 2 percent in the wood. Glue line temperature shall be determined by use of a potentiometer with a thermocouple located at the center of the innermost glue line. At the end of the cure cycle, pressure shall be immediately removed.

(a) *Heating period*. The elapsed time required to raise the innermost glue line of the clamped assembly from the ambient assembly temperature condition to the specified cure temperature.

(b) *Cure period*. The elapsed time during which the specified glue line cure temperature is maintained.

(c) *Cooling period*. The elapsed time required to cool the glued assembly to room temperature.

4.6.6.1.6.1 *Straight beams*. The two straight beams assembled at an ambient temperature of 23° ± 2° C (73.4° ± 3.6° F.) shall be cured at a glue line temperature of 65.6°

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

± 2° C. (150° ± 3.6° F.) for the period of time recommended in the manufacturer's instruction (see 3.7 for cure period limitations)

4.6.6.1.7 *Conditioning.* The beams prepared as specified shall be placed in a conditioning chamber maintained at a temperature of 23 ± 2° C. (73.4 ± 3.6° F.) and a relative humidity range of 55 to 74 percent (preferable optimum near 66 percent), until tests are conducted. Unless a lesser period of conditioning is advocated in the manufacturer's use instruction, preparation and test of block shear specimens specified in 4.6.6.2, and preparation and subjection of specimens to the cyclic delamination test specified in 4.6.6.3 shall be accomplished between 24 and 48 hours after completion of cure period and removal of clamps.

4.6.6.1.8 *Test sections.* The laminated beams shall be dressed on the sides to a uniform width of 6 inches. Four inches shall be trimmed off both ends of each beam and discarded. The remaining trimmed beam shall be cut into six cross sections as shown on figure 1. Cross-sections 1, 3 and 5, of figure — 1 shall be utilized for conducting block shear tests in accordance with 4.6.6.2, and cross sections 2, 4 and 6 shall be utilized for conducting tests for resistance to delamination in accordance with 4.6.6.3

4.6.6.2 *Shear test by compression loading*

4.6.6.2.1 *Preparation of test specimens.* Cross sections 1, 3 and 5, of figure (1), consisting of five glue lines per cross section, shall be prepared for test as shown on figure 2. Care shall be taken in preparing the test specimens to make the loaded surfaces smooth and parallel to each other and perpendicular to the height. When sawing the bonded assembly, care shall be exercised to insure that the saw cuts extend to, but not beyond, the adhesive line. The width and height of the specimen at the adhesive line shall be accurately measured to the nearest 0.010 inch to determine the shear area.

4.6.6.2.2 *Apparatus.* Apparatus for the block shear test shall be specified in method 1031 of Standard FED-STD-175.

4.6.6.2.3 *Test procedure.* The blocks shear test shall be conducted in accordance with the procedure specified in method 1031 of Standard FED-STD-175.

4.6.6.2.4 *Results.* Results of shear strength and percent wood failure shall be averaged for each test block cross section. Also shall be calculated the average shear strength and percent wood failure for all cross sections. In the event any of the requirements of 3.8 are not met in any one beam, one preparation and retest of the particular beam will be permitted, in the event all requirements (including delamination requirements) of the retest beam are met, results of tests conducted on the original beam may be disregarded.

4.6.6.3 *Test for resistance to delamination.*

4.6.6.3.1 *Specimens.* Specimens 2, 4 and 6 for each white oak beam prepared in accordance with 4.6.6.1, shall be dressed to the full cross section of the beam and 3 inches in length. The end surfaces shall be sawed smooth with a suitable saw or may be sanded smooth. Care shall be taken to prevent overheating of end surfaces during sanding

4.6.6.3.2 *Test procedure*

4.6.6.3.2.1 *Vacuum pressure cycle.* The specimens specified in 4.6.6.3.1 shall be placed in an autoclave, weighted down, and water at a temperature of 65° to 80° F. shall be admitted into the autoclave in quantity sufficient to submerge the specimens completely. Thereafter a vacuum of 20 to 25 inches of mercury shall be drawn and held for 2 hours. The vacuum shall then be broken, and a pressure of 80 ± 5 p.s.i. shall be applied to the submerged specimens and held for 2 hours. The pressure shall then be released and a vacuum of 20 to 25 inches of mercury shall again be drawn and held for 2 hours while the specimens remain submerged. The vacuum

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

shall be released again and a pressure of 80 ± 5 p.s.i. applied to the submerged specimens and held for 2 hours The pressure in the autoclave shall be reduced to atmospheric pressure before the water is drained out, and shall remain at atmospheric pressure during the draining operation

4.6.6.3.2.2 *Drying period.* The specimens shall be dried for a period of 88 hours (3⅔ days) in air maintained at 80° to 85° F. and 25 to 30 percent relative humidity, and circulating at the rate of 500 ± 50 feet per minute. During the drying the specimens shall be placed at least 2 inches apart and with the end surfaces parallel to the stream of air.

4.6.6.3.2.3 *Duration of test.* The cyclical exposure test shall consist of 3 cycles of soaking and drying specified in 4.6 6.3.2.1 and 4.6.6.3.2.2 comprising a total test period of 12 days.

4.6.6.3.2.4 *Measurement of delamination.*[1] Measurement of delamination to the nearest 0.05 inch shall be made on all adhesive lines of both end (cross sectional) surfaces of the test specimen at the end of the last cycle of the drying period. The percent delamination of the adhesive lines shall be calculated by dividing the total length of delamination measured along all adhesive lines (as stated above) by the total length of all adhesive lines on both exposure faces of the specimen.

4.6.6.3.2.5 *Results.* The percent delamination of the adhesive lines in each cross section test specimen shall be calculated by dividing the total length of delamination measured along all adhesive lines (as stated above) by the total length of all adhesive lines on both exposure faces of the specimen. Also shall be calculated the average percent delamination for all test specimens in a beam. In the event the requirement of 3.9 is not met in any one beam, one preparation and retest of the particular beam will be permitted; in the event all requirements (including block shear requirements) of the retest beam are met, results of tests conducted on the original beam may be disregarded.

4.6.7 *Storage life test.* A suitable amount of adhesive shall be stored for the period required in 3.10 at 23° ± 2° C. (73.4° ± 3 6° F.) and thereafter tested to determine conformance with the requirements of this specification, except for 3.2.3, 3.2.4 and 3.2.5.

4.8 **Inspection of preparation for delivery.** Sample cans, pails, drums, and shipping containers shall be inspected to verify conformance to the requirements of section 5 herein

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level A or C as specified in the contract or order (see 6.2)

5.1.1.1 *Level A.*

5.1.1.1 *Adhesive (liquid).* Liquid adhesive shall be furnished in 1 pint or 1 gallon cans, 5 gallon pails, or 30 or 55 gallon drums as specified (see 6.2(e))

5.1.1.1.1 *Cans.* Cans shall conform to type V, class 2 of Specification PPP-C-96. Exterior plan B coating and side seam stripping shall be required.

5.1.1.1.2 *Five-gallon pails* The 5-gallon pail shall conform to type II of Specification PPP-D-760 Color shall be as specified in the contract or order

5.1.1.1.3 *Thirty-gallon drums.* The 30-gallon drums shall conform to Specification 37A of the code of Federal Regulations Interior coating of drums shall not affect or be affected by the packaged product.

[1] Delamination is a term used to express separation of wood surfaces at the glue lines When the separation is in the wood, even though very close to the glue line, it is termed wood failure or checking Magnification is often necessary to determine whether the failure is in the glue or wood. A feeler gage, 0.004 inch in thickness is convenient for probing into the glue line to determine if separation actually exists.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

5.1.1.1.4 *Fifty-five gallon drums.* The 55-gallon drums shall conform to type III of Specification PPP–D–729. Interior coating of drums shall not affect or be affected by the packaged product.

5.1.1.2 *Hardener* The hardener shall be furnished in containers of such capacity as to provide the correct quantity required for an individual container of adhesive. Containers shall conform to Specification PPP–C–96 and exterior plan B coating and side seam stripping shall be required.

5.1.2 *Level C* The adhesive and the hardener shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 *Level A* Cans shall be arranged and packed for overseas shipment in accordance with the appendix to Specification PPP–C–96. Five, thirty and fifty-five gallon containers will not require packing

5.2.2 *Level B.* Cans shall be arranged and packed for domestic shipment in accordance with the appendix to Specification PPP–C–96 Five, thirty and fifty-five gallon containers will require no packing

5.2.3 *Level C* The adhesive and the hardener shall be packed to insure acceptance by common carrier and safe delivery to destination at the lowest applicable rate. Shipment shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 **Marking.** In addition to the marking instructions specified in 3.12 and any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129

6. NOTES

6.1 Laminating. Use of these adhesives in laminating various species of wood is covered by separate laminate specifications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Grade required (see 1.2)

(c) Size of container required (see section 5).

(d) Quantity (adhesive should be purchased by net weight).

(e) Level of packaging and level of packing required (see 5.1 and 5.2).

(f) Whether lot acceptance tests for individual orders of 15 gallons of adhesive or less will be waived (see 4.4.2).

(g) When a use instruction is not required for each unit container (see 3.11).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL–22397, whether or not such products have actually been so listed by that date The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the qualified products list is the Chief of the Bureau of Ships Department of the Navy Washington 25, D C., and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

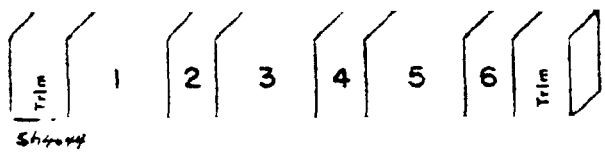
6.4 **Caution.** The user is cautioned that because of the formaldehyde content of the adhesive the usual cleanliness standards should be employed in its use. Where the users actually have adhesive coming in contact with the skin, it has been found that the use of protective creams and frequent washing alleviates any tendency toward industrial dermatitis unless, of course, the particular user is allergic to formaldehyde. It should be emphasized that many industrial dermatitis problems can be solved by proper cleanliness on the part of the worker and by the use of adequate plant ventilation.

6.5 **Supersession data.** This specification supersedes type II, class 2 of Specification MIL-A-397B dated 3 February 1953.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy—Bureau of Ships
(Project 8040-0028)

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ships
Air Force



A - STRAIGHT BEAM

FIGURE 1. *Laminated test beams*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.

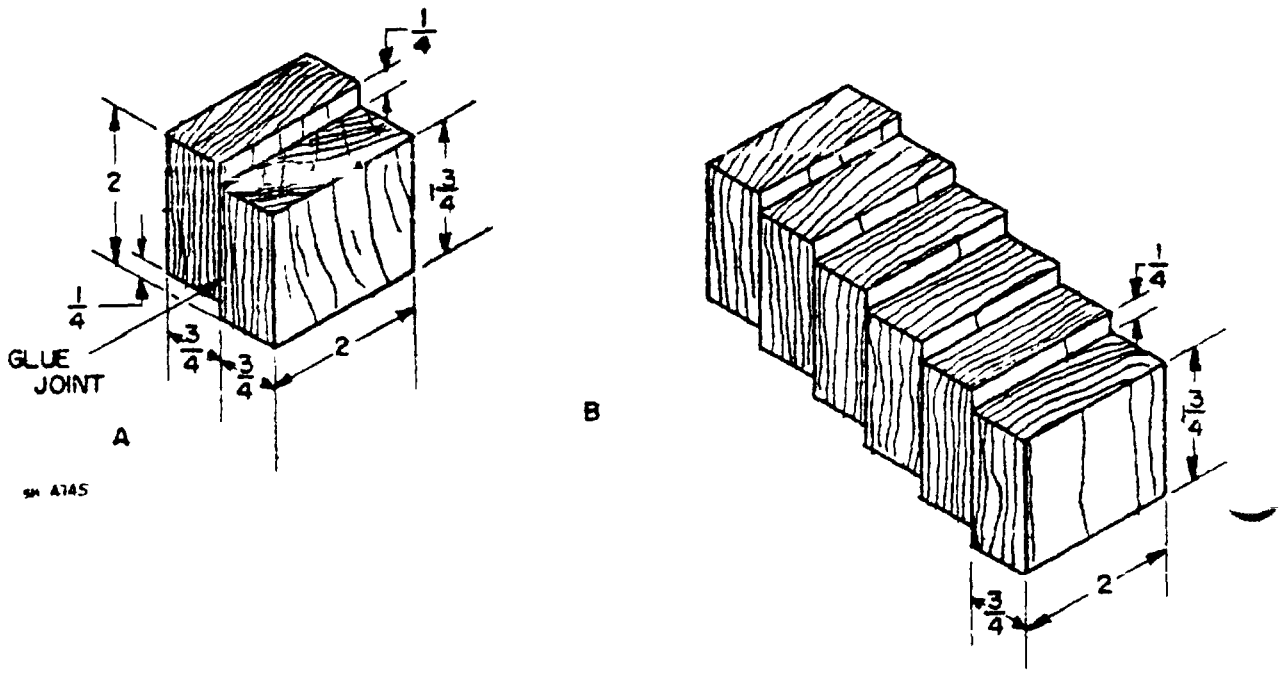


FIGURE 2 *Standard block (A) and stair step type (B) shear specimens for testing glue joint strength*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:13Z
Check the source to verify that this is the current version before use.